Emily Qi

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Education

University of Waterloo 2024 – 2028

Candidate for Bachelor of Computer Science, Honours (3.9 GPA)

- Computing and Financial Management Outstanding Achievement Award

Technical Skills

Languages: Python, C++, C, Java, JavaScript, TypeScript, SQL

Libraries and Frameworks: PyTorch, HuggingFace Transformers, OpenCV, Open3D, GTSAM, scikit-learn, ONNX, NumPy, pandas, ROS 2, FastAPI, React, Next.js, PostgreSQL

Tools: Git, Docker, Gazebo, Foxglove, COLMAP, Google Cloud, Weights & Biases

Experience

Machine Learning Engineering Intern Jan 2026 – Apr 2026

HubHead

- Built visual grounding validation across 250+ document types with Gemini 2.5 Pro, iteratively refining bounding box predictions via IoU against ground-truth for per-edge spatial feedback
- Designed multimodal template recommender with 768-dim Gemini embeddings, clarity-weighted statistical ranking, and persistent embedding caching for <1.0s inference across all user-generated templates
- Scaled ML extraction pipeline to 70K+ pages across Document AI, Gemini, and Textract on Google Cloud Run, offloading processing to a dedicated worker service with semaphore-gated concurrency and heartbeat fault recovery

Software Engineering Intern May 2025 – Aug 2025

Royal Canadian Mounted Police

- Drove complete development of full-stack app (React, Next.js, FastAPI, SQLite) with RESTful APIs and scheduled background jobs for operational workflow automation, reducing manual overhead by 70%
- Reverse-engineered bot-resistant web interfaces with Selenium for automated multi-platform interactions

Projects

SLAM from Scratch | *Python, OpenCV, GTSAM, scikit-learn*

- Built stereo visual odometry from scratch in Python using ORB features, KLT optical flow, and PnP-RANSAC; achieved 1.27% relative pose error on KITTI Odometry sequence 05
- Reduced trajectory drift by 73% by implementing bag-of-words loop closure (TF-IDF retrieval + PnP geometric verification) and GTSAM pose graph optimization
- Currently extending pipeline with IMU attitude integration using KITTI Raw 100Hz inertial data

Depth Estimation Benchmark | *Python, PyTorch, OpenCV, HuggingFace Transformers, ONNX, W&B*

- Benchmarked stereo and monocular neural depth estimation on 200 KITTI scenes; fine-tuned DPT-Large with Huber loss, improving AbsRel by 19% and surpassing the stronger pretrained DepthAnything V2 on MAE/RMSE
- Engineered classical stereo pipeline - calibration parsing, SGBM disparity tuning, and disparity-to-depth conversion - with bird's-eye-view occupancy map via 3D pinhole projection and height-filtered grid construction
- Exported fine-tuned DPT-Large to ONNX by patching ViT source to resolve dynamic tensor shapes; applied INT8 quantization achieving 4× model size reduction (1368MB → 352MB)

Autonomous Robot Navigation Stack | *C++, ROS 2, Gazebo, Docker, Foxglove*

- Built 4 C++/ROS 2 nodes (perception, mapping, planning, control) for a differential-drive robot with 2D lidar in Gazebo, implementing A* path planning on occupancy grids and global mapping via 2D rigid-body transforms
- Developed a pure-pursuit controller with rotate-in-place behavior and map-triggered replanning, cutting reaction latency from 500ms to sub-millisecond

Atari Breakout with DQN | *Python, PyTorch, Deep Reinforcement Learning, OpenCV, Weights & Biases*

- Combined three RL papers (DQN, Double DQN, Prioritized Experience Replay) from scratch in PyTorch; trained agent for 10M frames for 7× reward improvement and 227× Q-value improvement over random baseline
- Built memory-efficient prioritized replay buffer with uint8 frame compression (4× storage reduction), TD-error prioritization, and importance sampling bias correction for stable training on constrained GPU hardware